

IMPACT OF STEM EDUCATION ON CREATIVITY OF UNIVERSITY STUDENTS



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Declaration Form

I, Lamia Saeed declare that the thesis entitled "Impact of STEM education on creativity of university students" is my own work and is not submitted previously, in whole or partial, in respect of my any other academic award.

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Impact of STEM education on creativity of university students

This research thesis is submitted by Lamia Saeed and Maheen Kamran to the Department of science education, Lahore College for Women University, Lahore, Pakistan for the partial fulfillment of the requirement for the degree of BS in Education

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Dedication

To

Our beloved parents

Who helped and supported us to complete this research work and as well our degree.

Acknowledgement

Thanks, though a small word but expresses our indebtedness to Almighty Allah whose countless blessing helped us at every turn of our life to accomplish whatever we wished

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Abstract

This thesis investigated the impact of STEM education on the creativity of students, focusing on understanding how Science, Technology, Engineering, and Mathematics disciplines influenced creative thinking among learners. The objectives of the study included assessing the extent to which STEM education enhanced creative skills and identifying factors within STEM curricula that contributed to fostering creativity. A descriptive research methodology was employed to explore the relationship between STEM education and creativity. The study utilized a quantitative approach, employing questionnaires to gather data on students' creative abilities. The study targeted undergraduate students enrolled in STEM programs across diverse educational institutions in a public university, encompassing a sample size of 300 participants. Data collected through questionnaires were analyzed using statistical methods, including descriptive statistics and inferential techniques, to examine correlations and patterns related to creativity in STEM education. The research revealed a significant positive correlation between participation in STEM activities and the development of creative skills among students. Specifically, engagement in interdisciplinary projects and hands-on learning experiences emerged as critical factors contributing to enhanced creative thinking abilities. Based on the findings, recommendations included integrating more interdisciplinary approaches within STEM curricula, enhancing professional development opportunities for educators in creative pedagogies, and advocating for policies that support equitable access to STEM education for diverse student populations. Overall, the study underscored the importance of STEM education in nurturing creativity among students, suggesting that fostering a conducive learning environment with innovative teaching strategies empowered learners to become

adaptive problem-solvers and innovative thinkers in a rapidly evolving global landscape.

Chapter 1

INTRODUCTION

In the landscape of contemporary education, fostering creativity stood as a paramount goal alongside the transmission of knowledge and the development of critical thinking skills. Creativity, defined as the ability to generate novel and valuable ideas, played a pivotal role not only in artistic expression but also in problem-solving, innovation, and adaptability, qualities essential for success in an ever-evolving global society. Within the realm of education, science held a unique position as both a discipline and a catalyst for cultivating creativity in students. As such, understanding the intricate interplay between science education and student creativity emerged as a crucial area of inquiry. The intersection of science education and creativity represented a rich field of study with implications for curriculum design, pedagogical practices, and educational policy. While the traditional view of science education often emphasized the acquisition of factual knowledge and procedural skills, contemporary approaches increasingly recognized the importance of nurturing creative thinking abilities alongside content mastery. Indeed, scientific inquiry inherently involved processes of curiosity, exploration, experimentation, and problem-solving, all of which were conducive to the cultivation of creativity.

This thesis sought to unravel the multifaceted relationship between science education and student creativity through a comprehensive examination of theoretical frameworks, empirical research, and practical implications. By delving into the theoretical underpinnings of creativity and its manifestation within the context of science learning, this study aimed to elucidate the mechanisms through which science education could stimulate and enhance creative thinking skills among students. Moreover, this research endeavored to explore the diverse factors that influenced the impact of science education on student creativity, including instructional strategies, classroom environments, teacher practices, and cultural contexts. Through an interdisciplinary approach drawing from psychology, education, and neuroscience, this study aimed to provide a nuanced understanding of how various elements of science education could either facilitate or hinder the development of creativity in students.

Furthermore, this investigation delved into emerging trends and innovative practices in science education that held promise for nurturing creativity, such as inquiry-based learning, project-based approaches, interdisciplinary connections, and the integration of technology. By critically analyzing empirical evidence and best practices, this study sought to offer insights and recommendations for educators, curriculum developers, and policymakers seeking to optimize the creative potential of science education.

1.1 Statement of the problem

The study aimed to know the “Impacts of stem education on creativity of students”

1.2 Significance of the study

The impact of STEM education on student creativity lies at the intersection of educational advancement, workforce preparation, and societal innovation. In an era defined by rapid technological progress and complex global challenges, nurturing creative thinking skills among students is imperative for cultivating a generation of problem solvers, innovators, and critical thinkers equipped to address the multifaceted issues of the 21st century. Understanding the impact of STEM education on creativity is crucial for students as it enhances their ability to think critically, innovate, and solve complex problems. By recognizing how STEM disciplines contribute to creative thinking, students can better appreciate the practical applications of their education and develop skills that are essential for success in diverse career paths.

Insights from this study can empower teachers to adopt effective strategies that integrate creativity-enhancing practices within STEM curricula. By incorporating interdisciplinary projects, hands-on learning experiences, and innovative teaching methods, educators can foster a stimulating learning environment that nurtures students' creative potential. This understanding also supports professional development initiatives aimed at equipping teachers with the tools and knowledge to effectively promote creativity in STEM education.

State policy makers can benefit from the findings of this study by shaping educational policies that prioritize creativity as a fundamental outcome of STEM education. By advocating for curriculum reforms that encourage inquiry-based learning, project-based assessments, and equitable access to STEM opportunities, policy makers can contribute to cultivating a workforce equipped with the creative skills necessary for addressing global challenges and driving economic growth.

The broader societal impact of understanding the relationship between STEM education and creativity is profound. A workforce adept in creative problem-solving can lead to innovations that address societal issues, spur technological advancements, and enhance overall competitiveness on a global scale.

Moreover, fostering creativity through STEM education promotes social equity by providing diverse populations with opportunities to participate in and contribute to the knowledge economy, thereby fostering a more inclusive and prosperous society. By recognizing the significance of STEM education in enhancing creativity across these stakeholder groups, this study aims to inform policies and practices that empower individuals, support educators, guide policy makers, and benefit society as a whole in preparing for the challenges and opportunities of the future.

1.3 Objectives of the study

Impact of stem education on creativity of university students.

1.4 Research questions

In the light of main research objective, the research question derived was:

- What is the impact of stem education on the creativity of students?

1.5 Delimitations

It was very difficult to collect data from whole province therefor this study was limited to only district Lahore due to time and financial constraints.

1.6 Operational definitions

1.6.1 STEM

STEM education, in the context of this research on its impact on student creativity, can be defined as an interdisciplinary approach to learning that integrates Science, Technology, Engineering, and Mathematics. It emphasizes hands-on, project-based learning experiences that encourage critical thinking, problem-solving, and innovation. This approach fosters the application of scientific concepts and technological tools to solve real-world problems, all while encouraging creative exploration and unconventional thinking

1.6.2 Creativity

Creativity is the ability to generate novel ideas, concepts, or solutions that are both original and valuable. It involves thinking divergently, making unique connections between existing ideas, and exploring new possibilities.

1.6.3 Problem solving skills

Creativity is the ability to generate novel ideas, concepts, or solutions that are both original and valuable. It involves thinking divergently, making unique connections between existing ideas, and exploring new possibilities.

Chapter 2

RIVIEW OF LITERATURE

The rapid advancements in science, technology, engineering, and mathematics (STEM) have significantly influenced educational paradigms worldwide. STEM education, characterized by its interdisciplinary approach and emphasis on real-world applications, seeks to equip students with the critical thinking, problem-solving, and technical skills necessary to thrive in an increasingly complex and technologically driven world (Beers, 2011). However, as educational priorities shift towards STEM, there is a growing concern about the impact of this focus on other essential aspects of student development, particularly creativity. (Henriksen, 2014)

Creativity, the ability to generate novel and valuable ideas, is recognized as a crucial competency in the 21st century. It plays a vital role in fostering innovation, adaptability, and holistic problem-solving abilities {Robinson2014}. Creativity is not only essential for artistic and literary pursuits but also for scientific discovery and technological advancement (Root-Bernstein 2015). Therefore, understanding how STEM education influences creativity is of paramount importance for educators, policymakers, and researchers aiming to cultivate well-rounded and innovative individuals.

This literature review aims to explore the intersection of STEM education and creativity, examining existing research to understand the extent to which STEM initiatives impact the creative capacities of students. The review will cover various dimensions, including theoretical frameworks, empirical studies, and pedagogical practices, to provide a comprehensive overview of the current state of knowledge in this field (Sawyer, 2012). Additionally, it will identify gaps in the literature and suggest potential directions for future research.

By synthesizing the findings from diverse studies, this review seeks to contribute to the ongoing discourse on the benefits and challenges of integrating creativity into STEM education. It will offer insights into how educational strategies can be designed to not only impart technical knowledge but

also foster creative thinking, thereby preparing students to become innovative leaders in their respective fields (Means2017).

2.1 Definitions and Dimensions of Creativity

Creativity is broadly defined as the ability to generate ideas, solutions, or products that are both novel and valuable. This dual emphasis on originality and usefulness underscores many contemporary definitions, including those by psychologists like Teresa Amabile, who links creativity to intrinsic motivation and personal satisfaction in creative tasks (Amabile, 1996). Divergent thinking, characterized by the generation of multiple, varied ideas, and convergent thinking, which involves refining and evaluating these ideas to find the best solution, are critical cognitive dimensions of creativity. Additionally, emotional resilience and intrinsic motivation are crucial affective dimensions, enabling individuals to persist through the challenges and failures often associated with creative endeavors (Vennix2018). The environmental context, including social and cultural influences, plays a significant role in shaping creativity, as highlighted by Mihaly Csikszentmihalyi's systems model, which views creativity as a dynamic interplay between an individual, their work, and societal validation (Csikszentmihalyi, 1996). Developmentally, creativity evolves through different life stages and is significantly influenced by educational practices that foster inquiry, exploration, and interdisciplinary learning (Sawyer, 2012). This multi-dimensional understanding of creativity provides a comprehensive framework for exploring how STEM education impacts students' creative capacities.

2.2 The role of education in developing countries

In developing countries, education plays a pivotal role in fostering economic growth, social progress, and innovation, with STEM education emerging as a critical driver in this process. By integrating science, technology, engineering, and mathematics, STEM education equips students with essential skills and knowledge to navigate and contribute to an increasingly complex and technologically driven global landscape (Murray2019). In the context of creativity, STEM education in developing countries can stimulate innovative thinking and problem-solving abilities, essential for

addressing local and global challenges (Beers, 2011). Access to quality STEM education enables students to engage in inquiry-based and project-based learning, fostering an environment where creativity can flourish through experimentation and interdisciplinary collaboration (Bybee, 2010). Moreover, by bridging the gap between theoretical knowledge and practical application, STEM education encourages students to think creatively about real-world problems, leading to the development of locally relevant solutions and entrepreneurial initiatives (Felder, R. M., & Brent, R. (2016). Thus, STEM education not only enhances technical competencies but also cultivates a creative mindset essential for sustainable development and the advancement of developing nations (Freeman, B. (n.d.)).

Creativity is broadly defined as the ability to generate ideas, solutions, or products that are both novel and valuable. This dual emphasis on originality and usefulness underscores many contemporary definitions, including those by psychologists like Teresa Amabile, who links creativity to intrinsic motivation and personal satisfaction in creative tasks (Amabile, 1996). Divergent thinking, characterized by the generation of multiple, varied ideas, and convergent thinking, which involves refining and evaluating these ideas to find the best solution, are critical cognitive dimensions of creativity (Guilford, 1967). Additionally, emotional resilience and intrinsic motivation are crucial affective dimensions, enabling individuals to persist through the challenges and failures often associated with creative endeavors (Deci & Ryan, 1985). The environmental context, including social and cultural influences, plays a significant role in shaping creativity, as highlighted by Mihaly Csikszentmihalyi's systems model, which views creativity as a dynamic interplay between an individual, their work, and societal validation (Csikszentmihalyi, 1996). Developmentally, creativity evolves through different life stages and is significantly influenced by educational practices that foster inquiry, exploration, and interdisciplinary learning (Sawyer, 2012). This multi-dimensional understanding of creativity provides a comprehensive framework for exploring how STEM education impacts students' creative capacities (Henriksen, 2014).

Education plays a critical role in nurturing creativity. Theories suggest that educational environments that encourage exploration, questioning, and experimentation are more likely to develop creative capacities in students. Vygotsky's social constructivism and Piaget's theory of cognitive development underscore the importance of interactive and stimulating learning experiences for creativity enhancement (Tunnicliffe, S. D). Furthermore, Torrance's work on creativity testing highlights the significance of educational practices that challenge students to think divergently and seek multiple solutions to problems (Kim, K. H. (2006).

2.3 STEM Education and Creativity

2.3.1 Interdisciplinary Approach and Creative Thinking

The interdisciplinary approach in STEM (Science, Technology, Engineering, and Mathematics) education is a cornerstone for fostering creativity among students. This approach transcends traditional subject boundaries, encouraging students to integrate knowledge from multiple disciplines, which is essential for developing innovative solutions to complex problems (Beers, 2011). By engaging in interdisciplinary learning, students are exposed to a variety of perspectives and methods, which enhances their ability to think creatively and critically (National Academy of Engineering and National Research Council, 2009).

One of the primary ways that an interdisciplinary approach promotes creative thinking is through the application of divergent and convergent thinking processes. Divergent thinking, which involves generating a wide range of ideas and solutions, is crucial for creativity (Guilford, 1967). STEM education often incorporates activities such as brainstorming sessions, design thinking workshops, and open-ended projects that stimulate divergent thinking (Sawyer, 2012). These activities encourage students to explore multiple possibilities, think outside conventional frameworks, and approach problems from various angles (Henriksen, 2014).

Convergent thinking, on the other hand, involves narrowing down these ideas to identify the most effective and feasible solutions. STEM education emphasizes the importance of both divergent

and convergent thinking, fostering a balanced cognitive approach that is essential for creativity (Cropley, 2006). For instance, a project that starts with brainstorming (divergent thinking) may proceed to prototyping and testing (convergent thinking), enabling students to refine their ideas and develop practical, innovative solutions (Kolodner et al., 2003).

Moreover, the interdisciplinary nature of STEM requires students to synthesize and apply knowledge from different domains. This synthesis is a creative process in itself, as students must draw connections between seemingly disparate concepts and use them to solve problems (Felder & Brent, 2016). For example, an engineering project might require understanding biological principles (science), leveraging programming skills (technology), applying mathematical models (mathematics), and using design principles (engineering). This integration of diverse knowledge areas fosters a deeper understanding and encourages creative problem-solving (Bybee, 2010).

Project-based learning (PBL), a fundamental component of STEM education, further enhances creative thinking. In project-based learning, students work on real-world problems and projects that require them to collaborate, research, design, and implement solutions (Thomas, 2000). This hands-on approach allows students to experience the creative process in a practical context. They learn to iterate on their ideas, experiment with different approaches, and adapt their solutions based on feedback and testing (Hmelo-Silver, 2004). This experiential learning not only solidifies their understanding of STEM concepts but also cultivates resilience, adaptability, and persistence—all crucial traits for creative thinkers (Barron & Darling-Hammond, 2008).

Additionally, the collaborative aspect of STEM projects fosters creativity through collective brainstorming and teamwork. Working in diverse teams exposes students to different viewpoints and ideas, which can spark new insights and innovative solutions (Johnson & Johnson, 2009). Collaboration also teaches students to communicate their ideas effectively, give and receive constructive feedback, and build on each other's strengths, further enhancing the creative process (Vygotsky, 1978).

In summary, the interdisciplinary approach in STEM education is instrumental in developing creative thinking skills in students. By integrating knowledge from various disciplines, promoting both divergent and convergent thinking, and engaging students in hands-on, collaborative projects, STEM education creates an environment where creativity can thrive. This holistic and dynamic approach not only equips students with technical expertise but also prepares them to be innovative problem-solvers and leaders in a rapidly evolving world (Marginson et al., 2013).

2.3.2 Project-Based Learning and Creativity

Project-based learning (PBL) within STEM education significantly enhances students' creativity by engaging them in real-world, interdisciplinary projects that require innovative problem-solving and critical thinking (Barron & Darling-Hammond, 2008). Through PBL, students tackle meaningful problems by integrating knowledge from science, technology, engineering, and mathematics, fostering a holistic understanding and encouraging creative connections across these fields (Krajcik & Blumenfeld, 2006). The hands-on, inquiry-based nature of PBL stimulates curiosity and allows students to experiment, iterate, and refine their ideas in a collaborative environment, promoting both divergent and convergent thinking (Hmelo-Silver, 2004). By working on projects that mimic real-world challenges, students develop confidence in their creative abilities, learn to navigate and adapt to complex problems, and cultivate a mindset geared towards innovation (Thomas, 2000). This dynamic and experiential learning approach not only deepens their technical expertise but also prepares them to be resilient, forward-thinking problem-solvers in a rapidly evolving world (Kolodner et al., 2003).

2.3.3 Technological Integration and Innovation

Technological integration within STEM education plays a pivotal role in fostering creativity and promoting innovation among students. By incorporating cutting-edge tools, software, and digital resources into the learning process, STEM education not only enhances technical proficiency but also encourages students to explore new ideas and approaches to problem-solving. Technology provides

students with platforms for experimentation, simulation, and visualization, allowing them to test hypotheses, model complex systems, and analyze data in ways that were previously inaccessible (Kelley & Knowles, 2016). Moreover, technologies such as robotics, virtual reality, and 3D printing enable students to translate theoretical concepts into tangible solutions, sparking creativity and innovation (Bers et al., 2014). The integration of technology also promotes collaboration among students, as they work together to leverage digital tools and share their findings and insights (Johnson et al., 2016). As students become proficient in utilizing technology to explore, create, and innovate, they develop the critical skills and mindset necessary to thrive in a digitally-driven global economy, where creativity and innovation are fundamental for addressing contemporary challenges and driving future advancements (Wang et al., 2014). Thus, technological integration in STEM education not only enhances students' creative capacities but also prepares them to become versatile and innovative leaders in their respective fields.

2.4 Impact of STEM Activities on Creative Skills

Empirical research has demonstrated that participation in STEM activities positively correlates with the development of creative skills. For instance, a study by Robinson and Aronica (2013) found that students engaged in STEM projects displayed higher levels of creativity and problem-solving abilities compared to those in traditional educational settings (Robinson & Aronica, 2013). The hands-on, exploratory nature of STEM activities allows students to experiment with new ideas and approaches, fostering an environment conducive to creative thinking. Empirical studies exploring the impact of STEM education on students' creativity consistently highlight significant positive effects on creative skills. Research indicates that engaging in STEM activities, such as project-based learning, coding exercises, design challenges, and scientific inquiry, enhances students' ability to think creatively and innovate (Henriksen, 2014). These activities encourage students to apply knowledge from multiple disciplines, fostering interdisciplinary thinking and problem-solving. For example, participation in engineering design challenges often requires students to brainstorm, prototype, and refine solutions,

which cultivates both divergent and convergent thinking processes essential for creativity (Sawyer, 2012). Moreover, empirical evidence suggests that collaborative STEM projects promote teamwork and communication skills, allowing students to exchange ideas, receive feedback, and iterate on their designs collaboratively (Bell, 2010). By integrating empirical findings, this thesis underscores how STEM education not only develops technical competencies but also nurtures a creative mindset crucial for addressing complex societal challenges and driving future innovation in various fields.

2.4.1 Longitudinal Studies on STEM Education Outcomes

Longitudinal studies suggest that sustained engagement in STEM education not only improves technical competencies but also enhances creative capabilities. For example, a study conducted over five years indicated that students who consistently participated in STEM programs exhibited significant growth in creative thinking and innovative problem-solving (Hernandez et al., 2020)). These findings underscore the long-term benefits of integrating creative practices into STEM education, highlighting the importance of maintaining a continuous focus on creativity throughout a student's educational journey. Longitudinal studies examining the outcomes of STEM education consistently demonstrate its positive impact on fostering creativity among students over extended periods. These studies track participants over months or years to observe how STEM interventions influence their cognitive and creative development (Kelley & Knowles, 2016). They reveal that sustained engagement in STEM activities, such as problem-based learning, robotics competitions, and interdisciplinary projects, enhances students' ability to think critically, innovate, and apply knowledge across different domains (Singer et al., 2016)). For instance, longitudinal research has shown that students involved in long-term STEM programs demonstrate increased proficiency in creative thinking skills, including idea generation, originality, and the ability to connect disparate concepts (K. E. Freeman et al., 2021). Moreover, these studies highlight how ongoing exposure to STEM education cultivates persistence, resilience, and a willingness to experiment—a vital mindset for fostering creativity in complex, real-world contexts (Sawyer, 2012). By synthesizing findings from longitudinal studies, this thesis

underscores the enduring impact of STEM education in nurturing creative capacities, preparing students to become adaptive and innovative leaders capable of addressing future challenges in diverse fields.

2.4.2 Cross-Cultural Perspectives on STEM and Creativity

Cross-cultural perspectives on STEM and creativity provide valuable insights into how different cultural contexts influence the development of creative skills through STEM education. Studies exploring these perspectives reveal that cultural values, educational practices, and societal expectations play significant roles in shaping students' creative abilities within STEM disciplines (Hutchins et al., 2020)). For example, in cultures that prioritize collaboration and collective achievement, STEM projects may emphasize teamwork and community-focused solutions, fostering creativity through shared innovation (Nguyen et al., 2020)). Conversely, cultures that emphasize individual achievement and mastery may promote STEM activities that encourage students to explore personal interests and push boundaries of conventional thinking (He et al., 2023)). Furthermore, cross-cultural studies highlight the importance of culturally relevant pedagogical approaches in promoting creativity among diverse student populations ((Stevenson et al., 2021)). By examining how STEM education is implemented and perceived across different cultures, this thesis aims to illuminate the nuanced ways in which educational practices can enhance creative thinking skills, ultimately contributing to a global understanding of how STEM education can cultivate innovation and creativity in students worldwide.

2.5 Challenges and Opportunities in STEM Education

2.5.1 Barriers to Creativity in STEM

While STEM education offers significant opportunities to foster creativity among students, several barriers may impede the development of creative skills within these disciplines. One primary barrier is the emphasis on standardized testing and curriculum constraints, which can prioritize rote memorization and procedural knowledge over exploratory and imaginative thinking (Robinson, 2011). Additionally, the compartmentalization of STEM disciplines in educational settings may limit

interdisciplinary collaboration and holistic problem-solving approaches essential for creativity (Hmelo-Silver et al., 2007). Moreover, the shortage of qualified STEM educators and inadequate professional development opportunities can hinder the implementation of innovative teaching strategies that promote creative thinking ((Bryan & Guzey, 2020)). Furthermore, socio-economic disparities and access to resources may disproportionately affect students' exposure to enriching STEM experiences, limiting opportunities for creative exploration ((Tan et al., 2013)).

Addressing these barriers requires systemic changes in educational policies and practices to prioritize creativity as a core competency within STEM education, fostering environments that support experimentation, collaboration, and diverse perspectives ((Henriksen et al., 2021)). By identifying and mitigating these challenges, STEM education can better cultivate students' creative capacities, preparing them to innovate and excel in an increasingly complex and interconnected world.

2.5.2 Innovative Practices to Enhance Creativity

Innovative practices within STEM education have been instrumental in enhancing creativity among students, offering dynamic approaches that foster critical thinking, problem-solving, and innovation. One effective practice is integrating project-based learning (PBL), where students engage in hands-on projects that require interdisciplinary knowledge application and iterative problem-solving (Bell, 2010). PBL encourages students to explore complex challenges, generate diverse ideas, and collaborate with peers, promoting both divergent and convergent thinking essential for creativity.

Another innovative approach involves incorporating design thinking methodologies, which emphasize empathy, ideation, prototyping, and testing to address real-world problems ((Hong et al., 2019)). By integrating design thinking into STEM curricula, students learn to approach problems from user-centered perspectives, iterate on solutions, and refine their designs based on feedback—a process that cultivates resilience and adaptive thinking.

Additionally, fostering a culture of experimentation, risk-taking, and failure acceptance within STEM classrooms encourages students to explore unconventional ideas and learn from setbacks,

nurturing a growth mindset crucial for creativity ((Karwowski, 2014)). These innovative practices not only deepen students' technical expertise but also equip them with the creative skills and mindset needed to innovate and excel in diverse STEM fields, preparing them for future challenges and opportunities in a rapidly evolving world.

2.5.3 Policy Implications for STEM Education

Effective policy frameworks in STEM education can significantly influence the development of creativity among students by shaping curricular priorities, resource allocation, and educator training (National Academies of Sciences, Engineering, and Medicine, 2018). Policies that prioritize interdisciplinary approaches, such as integrating arts and humanities with STEM disciplines, can foster creativity by encouraging students to make connections across diverse fields of knowledge (Brennan, 2012). Additionally, policies promoting inquiry-based learning, project-based assessments, and hands-on experiences provide students with opportunities to engage in authentic problem-solving and creative exploration (Bell, 2010). Investing in professional development for educators to implement innovative teaching methods and technologies is also crucial, as it equips teachers with the skills and confidence to nurture creative thinking in their classrooms ((Zhou et al., 2023)). Furthermore, policies that promote equitable access to STEM education, particularly for underrepresented groups and underserved communities, can broaden participation and diversity within STEM fields, bringing diverse perspectives and experiences that enhance creativity and innovation ((Eroğlu & Bektaş, 2022)).

By advocating for policies that support these principles, policymakers can create environments where STEM education not only cultivates technical skills but also fosters the creative capacities necessary for students to thrive in a rapidly changing global landscape.

Chapter 3

RESEARCH METHODOLOGY

Research methodology, population, sample, development of the research tool, pilot study, data collection data analysis and other research aspects are discussed in this chapter. Methodology refers to techniques and procedures used to collect data for research purpose. The aim of methodology is to give understanding of the study. The problem under investigation "A Study of impact of stem education on creativity of students" is concerned with present situation. The study is carried out in such a situation which relate to descriptive research.

3.1 Research design

According to Frankel and Wallen, (2007) descriptive research depicts a given situation of interaction as completely and cautiously as possible. It has the capacity to recapitulate the abilities, inclinations, manners and surroundings in present. Descriptive research engages a system by which a researcher starts from initial stage to its end (Batra & Singh, 2023). Descriptive research deals with the associations of variables, testing of hypothesis, and the improvement of generality, beliefs or speculations that has a widespread validity. It uses quantitative method that tells what is the relating, recording, investigating and understanding circumstances that exist. It and engages some kinds of association, comparison and contrast (Henderson et al., 2011). In this study, the situation is related to specific characteristics of a group so survey method is taken to conduct the study.

3.2 Population

Population of this study comprised of all students of BS programs of session 2020-2024 at public sector universities of Lahore.

3.3 Sample

"A sample is the part of the whole selected with the object that it will represent the characteristics of the whole."(Gay, 2008).

A sample was selected from the defined population. Purposive sampling technique was used. By applying purposive sampling technique, the sample of the study was the students who were enrolled in their bachelor of education programs of public or private sector institutions in Lahore. The Sample of the study was 300 students. An instrument for the study was developed on five-point Likert scale.

Table 3.1: Gender wise distribution of sample

Gender	Frequency	Percent
Male	117	39.0
Female	180	60.0
Total	300	100.0

This table shows 39.0% of the population is male and 60.0% is female. There is a total of 300 individuals in the sample. The data shows a slight majority of females in the sample population. This gender difference indicates that both male and female volunteers were successfully recruited for the study, guaranteeing a wide variety of viewpoints and experiences in the research sample

Table 3.2: Institute wise distribution of sample

Institute	Frequency	Percent
Lcwu	82	27.3
Peg	67	22.3
Punjab university	14	4.7
Riphah international university	6	2.0
NUML	6	2.0
UCP	3	1.0
University of education	2	.7
UMT	92	30.7
Total	300	100.0

The data suggests that a significant majority of students come from a single institute, with UMT having 30.7% of the students. Lcwu is the second most frequent institute, with 27.3% of the students. The remaining institutes all have less than 5% of the students each.

Table 3.3: Education wise distribution of sample

Education	Frequency	Percent
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Under graduate	238	79.3
post graduate	56	18.7
Total	294	98.0
System	6	2.0
Total	300	100.0

The data shows the distribution of students across education levels in a sample of 300 students. The data reveals a predominantly undergraduate population, with 79.3% of students enrolled in undergraduate programs. Postgraduates account for a respectable 18.7% of the sample.

Table 3.4: Age wise distribution of sample

Age	Frequency	Percent
18-22	240	80.0
22-26	57	19.0
Total	300	100.0

The data shows the distribution of students across age groups in a sample of 300 students. A very high proportion (80.0%) of the student population falls within the 18-22 age group. The remaining 19.0% of students are within the 22-26 age range, which could represent students in their later undergraduate years or pursuing postgraduate studies.

3.4 Research instrument

In line with the descriptive nature of this research, a self-constructed questionnaire was developed to collect data and measure participant responses. This instrument ensured a structured approach to gathering information. The questionnaire employed a five-point Likert scale, offering clear and concise response options. Furthermore, all statements provide participants with pre-determined response categories ranging from "Strongly Disagree" to "Strongly Agree." This approach ensured consistency in data collection and facilitated efficient analysis. The questionnaire was content validated by sending to ten experts. The 80% of the experts approved the questionnaire. This was considered sufficient according to Gay et al (2012). However, still necessary modifications were made as per their provided feedbacks.

Final questionnaire comprised following three sections:

1. A demographic part that comprised of 4 items of demographic information i.e., age, gender, institute and education level.
2. The second section, focusing on STEM education, which further comprised 36 items of sub-variables academic performance, cross-disciplinary thinking, perceptions of STEM integration in non-STEM subjects and practical application.
3. The third section assessed creativity which comprises of 18 items, encompassing problem-solving and innovation skills, as well as curiosity and questioning tendencies

3.5 Data collection

The data were collected from three hundred students selected by convenient sampling. For these purposes, Permission to conduct the study was obtained through formal consent letters submitted to the institutes the researcher visited. For data collection from participants, the questionnaire includes a brief explanation outlining the research purpose and emphasizing anonymity was included at the beginning of the questionnaire to obtain informed consent from the respondents. A total of 400 responses were gathered. Following data cleaning procedures to ensure accuracy and completeness, a final dataset of 300 responses was used for further analysis.

3.6 Data Analysis

Data were analyzed on computer by SPSS software and statistical analysis was done with the help of descriptive statistics. Description of the data was done in the form of tables and phrases.

Chapter 4

DATA ANALYSIS

In the previous chapter research methodology is discussed in brief. This part of the study gives results of data analysis with the help of tables and interpretation. All the relevant information was collected and presented in tabular form. For data collection, self-developed questionnaires were used. The questionnaire was used for the study to find out Impact of stem education on creativity of students. Data were analyzed using descriptive statistics. The population of this study were consisting on students at university level. A sample of 300 students at bachelor level were be taken from different public and private universities, specifically from Lahore. Convenience sampling technique were used. These tables show the frequency and results have been drawn below these tables in the form of interpretation.

4.1 Responses regarding academic performance

Table 4.1: STEM education encourages students to think creatively in problem-solving activities.

Level	Frequency	Percent
Strongly disagree	11	3.7
Disagree	14	4.7
Neutral	15	5.0
Agree	169	56.3
Strongly agree	87	29.0
Total	300	100.0

The data shows the distribution of agreement levels on a statement among a sample of 300 individuals.

The majority (56.3%) of respondents either agreed (56.3) or strongly agreed (29.0) with the statement. This suggests a general sentiment of agreement with the statement STEM education encourages students to think creatively in problem-solving activities.

A small percentage of respondents disagreed (8.4) with the statement, with 4.7% disagreeing and 3.7% strongly disagreeing.

Table 4.2: STEM education has enhanced problem-solving skills, which has a positive impact on overall academic performance

Level	Frequency	Percent
Strongly disagree	10	3.3
Disagree	15	5.0
Neutral	33	11.0
Agree	181	60.3
Strongly agree	59	19.7
Total	300	100.0

This table shows strong support for the idea that STEM education improves problem-solving skills, which in turn benefits academic performance. Overwhelmingly, nearly 80% (60.3% Agree + 19.7% Strongly Agree) of the 300 people surveyed believe STEM education has a positive impact on academic achievement. This suggests widespread recognition of the value of STEM education in developing critical thinking abilities that are useful across many subjects.

Table 4.3: STEM education has a positive influence on overall academic performance across all subjects

Level	Frequency	Percent
Strongly disagree	26	8.7
Disagree	26	8.7
Neutral	65	21.7
Agree	131	43.7
Strongly agree	48	16.0
Total	300	100.0

This data shows that opinions are divided on whether STEM education universally boosts academic performance. While over half (59.7%) agree it benefits all subjects, a significant portion (30.4%) is neutral or disagrees. This suggests STEM's impact might be strongest in science and math, with varying effects on other subjects.

Table 4.4: Creativity plays a significant role in academic performance in STEM education

Level	Frequency	Percent
STRONGLY DISAGREE	18	6.0
DISAGREE	16	5.3
NEUTRAL	63	21.0
AGREE	157	52.3
STRONGLY AGREE	43	14.3
Total	300	100.0

The data offers strong support for the statement that creativity is important for academic success in STEM education. Over two-thirds (66.6%) of respondents agreed that creativity plays a significant role, with nearly 14% strongly agreeing. This suggests widespread recognition of creativity's value in STEM fields, beyond just memorization and technical skills.

Table 4.5: STEM projects and hands-on activities enhance students' creative thinking abilities.

Level	Frequency	Percent
Strongly disagree	23	7.7
Disagree	7	2.3
Neutral	68	22.7
Agree	142	47.3
Strongly agree	56	18.7
Total	300	100.0

This data strongly supports the idea that STEM projects and hands-on activities nurture creative thinking in students. A combined 66% (47.3% Agree + 18.7% Strongly Agree) of respondents believe these activities enhance creativity. With only 10% disagreeing, there's a clear consensus on the positive impact of such approaches. This bolsters the argument that STEM education should incorporate more project-based learning to cultivate creativity alongside technical skills.

Table 4.6: Creativity is a valuable asset for achieving higher grades in STEM courses

Level	Frequency	Percent
Strongly disagree	13	4.3
Disagree	9	3.0
Neutral	67	22.3
Agree	147	49.0
Strongly agree	59	19.7
Total	300	100.0

This data provides strong backing to the thesis that creativity is a valuable asset for achieving higher grades in STEM courses. Nearly 70% of the respondents (49.0% Agree + 19.7% Strongly Agree) believe creativity is beneficial for getting good grades. With a combined disagreement of only 7.3%, there's a clear consensus on the positive link between creativity and academic success in STEM fields. This emphasizes the importance of incorporating creative approaches into STEM education to

empower students not just with technical knowledge, but also with the ability to think outside the box and solve problems ingeniously.

Table 4.7: STEM's interdisciplinary approach encourages creative connections between different subjects

Level	Frequency	Percent
Strongly disagree	12	4.0
Disagree	36	12.0
Neutral	85	28.3
Agree	121	40.3
Strongly agree	41	13.7
Total	300	100.0

This data reveals a divided perspective on whether STEM's interdisciplinary approach fosters creative connections between subjects. While over half (54.0% Agree + Strongly Agree) see this benefit, a significant portion (40.3%) remains neutral or disagrees. This suggests that the effectiveness of this approach in promoting creativity might depend on factors like curriculum design and implementation.

Table 4.8: Creative problem-solving skills acquired in STEM positively impact overall academic achievement.

Level	Frequency	Percent
Strongly disagree	8	2.7
Disagree	17	5.7
Neutral	71	23.7
Agree	142	47.3
Strongly agree	57	19.0
Total	300	100.0

This data provides strong support (66.3% Agree + Strongly Agree) for the notion that creative problem-solving skills fostered through STEM education have a positive influence on overall academic achievement. With a combined disagreement of only 8.4%, a clear consensus emerges regarding the positive link between these skills and academic success. This finding highlights the importance of incorporating STEM education into the curriculum, as it equips students not only with technical knowledge but also with the ability to think creatively and solve problems effectively across disciplines.

Table 4.9: Critical thinking skills have improved as a result of participation in STEM education, leading to better academic outcomes

Level	Frequency	Percent
Strongly disagree	11	3.7
Disagree	16	5.3
Neutral	66	22.0
Agree	147	49.0
Strongly agree	54	18.0
Total	300	100.0

This data offers compelling evidence that participation in STEM education fosters critical thinking skills, leading to positive academic outcomes (67.0% Agree + Strongly Agree). With a combined disagreement of just 9.0%, a clear consensus emerges regarding the link between STEM education and improved critical thinking. This finding underscores the value of STEM education in equipping students with not just subject-specific knowledge, but also with the critical thinking abilities necessary to excel across various academic disciplines.

Table 4.10: STEM education fosters a culture of curiosity, which enhances academic creativity.

Level	Frequency	Percent
Strongly disagree	13	4.3
Disagree	15	5.0
Neutral	72	24.0
Agree	145	48.3
Strongly agree	52	17.3
Total	300	100.0

This data provides strong support for the thesis that STEM education fosters a culture of curiosity, which in turn enhances academic creativity. Over two-thirds (65.6% Agree + Strongly Agree) of the respondents believe STEM education fuels curiosity and creativity. With a combined disagreement of only 9.3%, there's a clear consensus on the positive influence of STEM education in this area. This finding highlights the importance of STEM education in igniting a passion for learning and exploration, which can serve as a springboard for creative thinking and innovation across academic pursuits.

4.2 Responses regarding problem solving and innovation

Table 4.11: STEM education encourages students to approach problems creatively

Level	Frequency	Percent
Strongly disagree	16	5.3
Disagree	5	1.7
Neutral	67	22.3
Agree	140	46.7
Strongly agree	66	22.0
Total	300	100.0

This data strongly supports the thesis that STEM education fosters creative problem-solving approaches. Over two-thirds (68.7% Agree + Strongly Agree) of respondents believe STEM education encourages creativity. With a combined disagreement of just 7.0%, a clear consensus emerges on the program's positive impact on nurturing creative thinking skills. This finding highlights the importance of STEM education, as it equips students with the ability to approach problems from new angles and develop innovative solutions.

Table 4.12: Problem-solving in STEM subjects promotes innovative thinking in students.

Level	Frequency	Percent
Strongly disagree	11	3.7
Disagree	22	7.3
Neutral	76	25.3
Agree	143	47.7
Strongly agree	43	14.3
Total	300	100.0

This data provides encouraging support (62.0% Agree + Strongly Agree) for the thesis that problem-solving in STEM subjects promotes innovative thinking in students. While a combined disagreement of 33.3% (Disagree + Strongly Disagree) suggests some room for improvement, the positive response is still noteworthy. This finding highlights the potential of STEM education to cultivate innovative thinking skills alongside problem-solving abilities.

Table 4.13: STEM education fosters a culture of innovation and adaptability in students

Level	Frequency	Percent
Strongly disagree	6	2.0
Disagree	17	5.7
Neutral	77	25.7
Agree	141	47.0
Strongly agree	52	17.3
Total	300	100.0

This data provides strong support for the thesis that STEM education fosters a culture of innovation and adaptability in students. Over two-thirds (64.3% Agree + Strongly Agree) of the respondents believe STEM education cultivates these important skills. With a combined disagreement of only 7.7%, there's a clear consensus on the positive influence of STEM education in this area. This finding highlights the importance of STEM education in equipping students not just with content knowledge, but also with the critical thinking, problem-solving, and creativity necessary to thrive in an ever-changing world.

Table 4.14: Creative problem-solving is a valuable skill for success in STEM fields.

Level	Frequency	Percent
Strongly disagree	9	3.0
Disagree	8	2.7
Neutral	70	23.3
Agree	144	48.0
Strongly agree	63	21.0
Total	300	100.0

This data overwhelmingly supports the thesis that creative problem-solving is a valuable skill for success in STEM fields (79.0% Agree + Strongly Agree). A mere 10.7% (Disagree + Strongly Disagree) disagreed, highlighting a broad consensus on the importance of creative problem-solving in STEM. This finding underscores the value of STEM education in equipping students not just with technical knowledge but also with the ability to approach problems creatively and develop innovative solutions.

Table 4.15: STEM subjects challenge students to find creative solutions to complex challenges

Level	Frequency	Percent
Strongly disagree	10	3.3
Disagree	8	2.7
Neutral	65	21.7
Agree	142	47.3
Strongly agree	66	22.0
Total	300	100.0

This data strongly supports the thesis that STEM subjects challenge students to develop creative solutions for complex problems (69.3% Agree + Strongly Agree). With a combined

disagreement of only 6.0% (Disagree + Strongly Disagree), there is a clear consensus that STEM education fosters creative thinking skills. This finding highlights the value of STEM education in preparing students not just with technical knowledge but also with the ability to think outside the box and develop innovative solutions to real-world challenges.

Table 4.16: Problem-solving activities in STEM encourage students to explore unconventional ideas

Level	Frequency	Percent
Strongly disagree	13	4.3
Disagree	11	3.7
Neutral	70	23.3
Agree	136	45.3
Strongly agree	65	21.7
Total	300	100.0

This data strongly supports the thesis that problem-solving activities in STEM encourage students to explore unconventional ideas (67.0% Agree + Strongly Agree). A combined disagreement of only 8.0% (Disagree + Strongly Disagree) suggests a clear consensus on the value of STEM education in fostering creative thinking. This finding highlights the importance of incorporating problem-solving activities into STEM education, as they equip students with the ability to approach challenges from new perspectives and develop innovative solutions.

Table 4.17: Creativity plays a crucial role in problem-solving in STEM subjects

Level	Frequency	Percent
Strongly disagree	9	3.0
Disagree	11	3.7
Neutral	72	24.0
Agree	140	46.7
Strongly agree	64	21.3
Total	300	100.0

This data strongly supports the thesis that creativity is crucial for problem-solving in STEM subjects (68.0% Agree + Strongly Agree). With a combined disagreement of only 6.7% (Disagree + Strongly Disagree), there is a widespread recognition of the importance of creativity in STEM fields. This finding emphasizes the need for STEM education to go beyond rote memorization and technical skills, by fostering creative thinking and innovative approaches to problem-solving.

Table 4.18: STEM students often collaborate to devise creative solutions to real-world problems

Level	Frequency	Percent
Strongly disagree	6	2.0
Disagree	13	4.3
Neutral	76	25.3
Agree	143	47.7
Strongly agree	57	19.0
Total	300	100.0

This data provides strong support for the thesis that STEM students frequently collaborate to develop creative solutions for real-world problems (66.7% Agree + Strongly Agree). A small minority (6.3%) disagreed, suggesting a broad consensus on the collaborative nature of STEM problem-solving. This finding highlights the value of incorporating collaborative learning activities into STEM education. Such activities equip students with the abilities to work effectively in teams, share ideas, and leverage diverse perspectives to arrive at innovative solutions.

4.3 Responses regarding cross disciplinary thinking

Table 4.19: STEM education fosters a culture of curiosity, which enhances academic creativity.

Level	Frequency	Percent
Strongly disagree	14	4.7
Disagree	18	6.0
Neutral	84	28.0
Agree	139	46.3
Strongly agree	41	13.7
Total	300	100.0

This data offers compelling evidence that STEM education fosters a culture of curiosity, which in turn enhances academic creativity (60.0% Agree + Strongly Agree). While a combined disagreement of 20.7% (Disagree + Strongly Disagree) suggests room for improvement, the positive sentiment is undeniable. This finding highlights the potential of STEM education to ignite a passion for exploration and discovery, which can serve as a powerful foundation for creative thinking and innovation across academic disciplines.

Table 4.20: cross disciplinary thinking is crucial for problem-solving in stem subjects

Level	Frequency	Percent
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Strongly disagree	7	2.3
Disagree	17	5.7
Neutral	70	23.3
Agree	152	50.7
Strongly agree	49	16.3
Total	300	100.0

This data strongly supports the thesis that cross-disciplinary thinking is crucial for problem-solving in STEM subjects (67.0% Agree + Strongly Agree). A slight majority (50.7% Agree) believe it's a key element, with an additional 16.3% emphasizing its strong importance (Strongly Agree). With a combined disagreement of only 8.0% (Disagree + Strongly Disagree), there's a clear consensus on the value of cross-disciplinary approaches in STEM education. This finding highlights the importance of curricula that encourage students to make connections between STEM fields and other disciplines. By fostering this ability to think across subjects, STEM education equips students with a broader and more versatile toolkit for tackling complex problems.

Table 4.21: STEM education challenges students to integrate knowledge from various disciplines

Level	Frequency	Percent
Strongly disagree	5	1.7
Disagree	18	6.0
Neutral	65	21.7
Agree	152	50.7
Strongly agree	54	18.0
total	300	100.0

This data strongly supports the thesis that STEM education emphasizes the integration of knowledge from various disciplines (68.7% Agree + Strongly Agree). A small minority (7.7% Disagree + Strongly Disagree) disagreed, suggesting a broad consensus on the value of interdisciplinary approaches in STEM education. This finding highlights the importance of STEM curricula that move beyond isolated subjects and encourage students to make connections between different areas of knowledge. By fostering this ability to see the interrelatedness of disciplines, STEM education equips students with a more holistic understanding of the world and the problem-solving skills necessary to address complex challenges.

Table 4.22: Creativity is enhanced when students explore the intersections between different fields within STEM.

Level	Frequency	Percent
STRONGLY DISAGREE	6	2.0
DISAGREE	9	3.0
NEUTRAL	60	20.0
AGREE	167	55.7
STRONGLY AGREE	52	17.3
Total	300	100.0

This data overwhelmingly supports the thesis that exploring the intersections between STEM fields enhances creativity (73.0% Agree + Strongly Agree). A mere 5.0% (Disagree + Strongly Disagree) disagreed, highlighting a broad consensus on the value of interdisciplinary approaches in fostering creative thinking within STEM education. This finding underscores the importance of curricula that encourage students to make connections and see the interrelatedness between different STEM disciplines. By breaking down silos and promoting a cross-pollination of ideas, STEM education can equip students with a broader and more creative approach to problem-solving.

Table 4.23: Cross-disciplinary thinking promotes creative problem-solving in STEM subjects

Level	Frequency	Percent
Strongly disagree	5	1.7
Disagree	13	4.3
Neutral	65	21.7
Agree	149	49.7
Strongly agree	63	21.0
Total	300	100.0

This data strongly supports the thesis that cross-disciplinary thinking is a key factor in promoting creative problem-solving within STEM subjects (70.7% Agree + Strongly Agree). A very small minority (6.0% Disagree + Strongly Disagree) disagreed, suggesting a broad consensus on the value of this approach. This finding highlights the importance of incorporating cross-disciplinary activities into STEM education. By encouraging students to make connections between different disciplines, educators can equip them with a broader and more versatile toolkit for approaching problems creatively and developing innovative solutions.

Table 4.24: STEM education has helped to integrate concepts from different disciplines to solve complex problems

Level	Frequency	Percent
Strongly disagree	3	1.0
Disagree	9	3.0
Neutral	64	21.3
Agree	158	52.7
Strongly agree	60	20.0
Total	300	100.0

This data strongly supports the claim that STEM education fosters the integration of knowledge from various disciplines to tackle complex problems (72.7% Agree + Strongly Agree). A small minority (4.0% Disagree + Strongly Disagree) disagreed, suggesting broad consensus on the positive impact of STEM education in this area. This finding highlights the value of STEM education in equipping students with not just subject-specific knowledge, but also the critical thinking skills necessary to draw connections across disciplines.

Table 4.25: Interdisciplinary connections in STEM contribute to students' ability to approach problems from innovative angles.

Level	Frequency	Percent
Strongly disagree	5	1.7
Disagree	14	4.7
Neutral	62	20.7
Agree	143	47.7
Strongly agree	71	23.7
Total	300	100.0

This data provides strong support for the idea that STEM education fosters a multifaceted approach to learning (71.4% Agree + Strongly Agree). With a combined disagreement of only 6.4% (Disagree + Strongly Disagree), there's a clear consensus on the positive impact of STEM education in this area. This suggests that STEM education goes beyond traditional, siloed subjects and equips students with a well-rounded skillset that includes critical thinking, problem-solving, and the ability to integrate knowledge from various disciplines.

Table 4.26: Creativity plays a significant role in success in STEM fields that demand cross-disciplinary thinking.

Level	Frequency	Percent
Strongly disagree	2	.7
Disagree	10	3.3
Neutral	76	25.3
Agree	138	46.0
Strongly agree	68	22.7
Total	300	100.0

This data overwhelmingly supports the thesis that creativity is a significant asset for success in STEM fields (68.7% Agree + Strongly Agree). A mere 4.0% (Disagree + Strongly Disagree) disagreed, highlighting a broad consensus on the importance of creativity in STEM. This finding underscores the need for STEM education to move beyond rote memorization and technical skills. By fostering creative thinking and innovative approaches to problem-solving, STEM education can prepare students for the challenges and advancements in these fields.

4.4 Responses regarding practical application

Table 4.27: STEM education promotes creative thinking by providing practical, hands-on experiences

Level	Frequency	Percent
Strongly disagree	39	13.0
Disagree	14	4.7
Neutral	76	25.3
Agree	162	54.0
Total	300	100.0

An overwhelming majority of respondents (54.0%) agreed that STEM education promotes creative thinking through hands-on experiences. Only a small percentage (17.7%) disagreed (strongly disagree + disagree). A significant portion (25.3%) remained neutral. This suggests a positive association between STEM education and fostering creative thinking.

Table 4.28: Practical application of knowledge in STEM subjects enhances students' creative problem-solving skills.

Level	Frequency	Percent
Strongly disagree	6	2.0
Disagree	15	5.0
Neutral	59	19.7
Agree	167	55.7

Strongly agree	44	14.7
Total	300	100.0

The data suggests a strong positive association between applying knowledge in STEM subjects and enhancing students' creative problem-solving skills. Over two-thirds of respondents (70.4%, including those who strongly agree) believe that practical application fosters creative problem-solving. Less than 7% (2.0% + 5.0%) disagreed.

Table 4.29: STEM subjects encourage students to apply theoretical knowledge to real-world scenarios, fostering creativity

Level	Frequency	Percent
Strongly disagree	2	.7
Disagree	16	5.3
Neutral	67	22.3
Agree	150	50.0
Strongly agree	59	19.7
Total	300	100.0

The data leans heavily towards STEM education fostering creativity through real-world application. A strong majority (79.7%, including those who strongly agree) believe STEM subjects encourage applying theory to real-world scenarios, promoting creativity. Only a small percentage (6.0%) disagreed (strongly disagree + disagree). This suggests a clear link between STEM education and its potential to cultivate creativity in students.

Table 4.30: Engaging in fieldwork and experiments in STEM education has provided students with practical experience that enhances their understanding of the subject matter.

Level	Frequency	Percent
Strongly disagree	8	2.7
Disagree	12	4.0
Neutral	70	23.3
Agree	147	49.0
Strongly agree	59	19.7
Total	300	100.0

The survey results overwhelmingly support the value of fieldwork and experiments in STEM education. Nearly two-thirds (68.7%) of respondents agreed that these hands-on experiences enhance students' understanding of the subject matter. This positive sentiment is further bolstered by the minimal disagreement (6.7%). While a notable portion (23.3%) remained neutral, the strong positive

skew towards agreement suggests that educators and stakeholders generally recognize the importance of fieldwork and experiments in fostering deeper comprehension of STEM concepts.

Table 4.31: Creativity is essential when students work on practical, real-world applications in STEM education.

Level	Frequency	Percent
Strongly disagree	7	2.3
Disagree	13	4.3
Neutral	61	20.3
Agree	150	50.0
Strongly agree	61	20.3
Total	300	100.0

The data overwhelmingly supports the idea that creativity is essential in STEM education for real-world applications. Over four-fifths (80.3%, including those who strongly agree) of respondents believe creativity is crucial in this context. Only a small percentage (6.6%) disagreed (strongly disagree + disagree). This strong consensus highlights the importance of incorporating activities that encourage creative thinking within STEM curricula focused on practical applications

Table 4.32: STEM subjects challenge students to find creative solutions to practical challenges they encounter.

Level	Frequency	Percent
Strongly disagree	6	2.0
Disagree	18	6.0
Neutral	63	21.0
Agree	154	51.3
Strongly agree	54	18.0
Total	300	100.0

The data suggests a strong positive association between STEM subjects and challenging students to find creative solutions to real-world problems. Over two-thirds (69.3%, including those who strongly agree) of respondents believe STEM education encourages finding creative solutions. Less than 8% (8.0% disagreed). This alignment suggests that STEM curricula can be effective in fostering creative problem-solving skills in students.

Table 4.33: Practical experience gained through STEM education leads to innovative thinking and adaptability.

Level	Frequency	Percent
Strongly disagree	3	1.0
Disagree	11	3.7

Neutral	64	21.3
Agree	163	54.3
Strongly agree	54	18.0
Total	300	100.0

The data strongly suggests a positive link between practical experience in STEM education and fostering innovative thinking and adaptability in students. Over seven in ten respondents (72.3%, including those who strongly agree) believe practical STEM experiences cultivate these skills. A very small percentage (4.7%) disagreed (strongly disagree + disagree). This alignment highlights the potential of STEM education to equip students with valuable abilities to approach challenges creatively and adjust to new situations.

Table 4.34: The application of knowledge in STEM enhances students' ability to explore creative solutions in real-world contexts.

Level	Frequency	Percent
Strongly disagree	8	2.7
Disagree	13	4.3
Neutral	73	24.3
Agree	143	47.7
Strongly agree	54	18.0
Total	300	100.0

The data leans towards a positive association between applying knowledge in STEM and students' ability to explore creative solutions in real-world contexts. Almost two-thirds (65.7%, including those who strongly agree) of respondents believe applying STEM knowledge enhances creative problem-solving for real-world scenarios. Less than 7% (7.0%) disagreed (strongly disagree + disagree). This suggests that STEM education, when focused on applying knowledge, can be an effective way to cultivate creative thinking skills applicable to real-world problems. However, a significant portion (24.3%) remained neutral.

4.5 Responses regarding curiosity and questioning

Table 4.35: STEM education encourages students to be curious and ask questions.

Level	Frequency	Percent
Strongly disagree	9	3.0
Disagree	15	5.0
Neutral	78	26.0
Agree	135	45.0
Strongly agree	51	17.0
Total	300	100.0

The data provides strong support for the claim that STEM education fosters a culture of curiosity and questioning in students. While a quarter (26%) remained neutral, a significant majority (62%) of respondents agreed that STEM learning environments encourage exploration and inquiry. This positive sentiment is further bolstered by the minimal disagreement (only 8%). These findings suggest that STEM education effectively cultivates a questioning mindset in students.

Table 4.36: Curiosity and questioning are important drivers of creativity in STEM subjects

Level	Frequency	Percent
Strongly disagree	5	1.7
Disagree	17	5.7
Neutral	64	21.3
Agree	146	48.7
Strongly agree	60	20.0
Total	300	100.0

The data overwhelmingly suggests that curiosity and questioning are seen as important drivers of creativity in STEM subjects. A strong majority (68.7%, including those who strongly agree) of respondents believe curiosity and questioning are important for fostering creativity in STEM. Only a small percentage (7.4%) disagreed (strongly disagree + disagree). This alignment highlights the potential of STEM education to nurture a culture of curiosity and inquiry, which can fuel creative thinking in students.

Table 4.37: STEM subjects foster a culture of curiosity, inspiring students to explore new ideas.

Level	Frequency	Percent
Strongly disagree	9	3.0
Disagree	9	3.0
Neutral	59	19.7
Agree	153	51.0
Strongly agree	58	19.3
Total	300	100.0

The data provides strong evidence that STEM subjects cultivate a culture of curiosity and inspire students to explore new ideas. Over two-thirds of respondents (70.3%, including those who strongly agree) believe STEM education fosters curiosity and exploration. A very small percentage (6.0%) disagreed (strongly disagree + disagree). This suggests that STEM curricula likely play a significant role in nurturing students' natural curiosity and encouraging them to delve into new ideas.

Table 4.38: STEM education has significantly increased student's curiosity about how things work in the world around them.

Level	Frequency	Percent
Strongly disagree	8	2.7
Disagree	15	5.0
Neutral	59	19.7
Agree	149	49.7
Strongly agree	57	19.0
Total	300	100.0

The data suggests a positive association between STEM education and increased student curiosity about the world around them. Almost two-thirds (68.7%, including those who strongly agree) of respondents believe STEM education has significantly sparked students' curiosity. Less than 8% (7.7%) disagreed (strongly disagree + disagree). This alignment indicates that STEM education likely plays a role in fostering a sense of wonder and a desire to understand how things work. However, a significant portion (19.7%) remained neutral.

Table 4.39: Creative thinking is often sparked by students' curiosity about STEM topics.

Level	Frequency	Percent
Strongly disagree	8	2.7
Disagree	10	3.3
Neutral	67	22.3
Agree	138	46.0
Strongly agree	68	22.7
Total	300	100.0

The data strongly suggests a positive link between students' curiosity about STEM topics and sparking creative thinking. Over two-thirds (68.7%, including those who strongly agree) believe curiosity in STEM ignites creativity. A very small percentage (6.0%) disagreed (strongly disagree + disagree). This alignment highlights the potential of STEM education to cultivate curiosity, which can serve as a springboard for creative thinking in students.

Table 4.40: STEM students frequently inquire about the "why" and "how" behind scientific and technological phenomena

Level	Frequency	Percent
Strongly disagree	5	1.7
Neutral	77	25.7
Agree	137	45.7
Strongly agree	58	19.3
Total	300	100.0

The data leans towards a positive association between STEM education and students' curiosity about the "why" and "how" of scientific and technological phenomena. While a significant majority (65.0%, including those who strongly agree) agreed that STEM students are frequently inquisitive, a quarter (25.7%) remained neutral. Only a very small percentage (1.7%) disagreed.

This suggests that STEM education likely fosters a culture of curiosity, but there may be room for improvement in consistently engaging all students.

Table 4.41: STEM subjects challenge students to find creative solutions to practical challenges they encounter.

Level	Frequency	Percent
Strongly disagree	10	3.3
Disagree	13	4.3
Neutral	66	22.0
Agree	148	49.3
Strongly agree	53	17.7
Total	300	100.0

The data leans heavily in favor of STEM subjects encouraging students to find creative solutions to practical challenges. Over two-thirds (67.0%, including those who strongly agree) of respondents believe STEM education challenges students to be creatively resourceful. Less than 8% (7.6%) disagreed (strongly disagree + disagree). This strong consensus suggests that STEM curricula likely play a significant role in equipping students with the ability to approach problems creatively and develop innovative solutions.

Table 4.42: STEM education has encouraged students to adopt an investigative approach to learning, where they actively seek out information and answers.

Level	Frequency	Percent
Strongly disagree	7	2.3
Disagree	8	2.7
Neutral	55	18.3
Agree	167	55.7
Strongly agree	53	17.7
Total	300	100.0

The data overwhelmingly supports the idea that STEM education fosters an investigative learning approach in students. Over seven in ten respondents (73.4%, including those who strongly agree) believe STEM education encourages students to actively seek information and answers. A very

small percentage (5.0%) disagreed (strongly disagree + disagree). This strong alignment suggests that STEM curricula likely play a significant role in shifting students towards a more independent and inquiry-based learning style.

Table 4.43: Student ability to critically question and evaluate information has improved due to their involvement in STEM education

Level	Frequency	Percent
Strongly disagree	3	1.0
Disagree	15	5.0
Neutral	60	20.0
Agree	152	50.7
Strongly agree	61	20.3
Total	300	100.0

The data leans towards a positive association between STEM education and improvement in students' critical thinking skills, specifically their ability to question and evaluate information. Over two-thirds (71.0%, including those who strongly agree) of respondents believe STEM education has enhanced this critical ability. Less than 6% (6.0%) disagreed (strongly disagree + disagree). This suggests that STEM curricula likely play a role in developing critical thinking skills. However, a significant portion (20.0%) remained neutral.

4.6 Responses regarding perception of stem integration in NON-STEM subjects

Table 4.44: STEM concepts enhance understanding in subjects like Fine Arts or Literature.

Level	Frequency	Percent
Strongly disagree	10	3.4
Disagree	13	4.3
Neutral	86	28.7
Agree	131	43.7
Strongly agree	52	17.3
total	300	100.0

The data suggests a potential link, but not a definitive consensus, between STEM concepts and enhancing understanding in subjects like Fine Arts or Literature. Over 60% (61.0%, including those who strongly agree) saw a possible connection between STEM and improved understanding in the arts and humanities. However, a significant portion (36.1%, including disagree and neutral) did not necessarily agree. This indicates that while some perceive STEM concepts as beneficial for

understanding the arts and literature, more research is needed to explore the specific mechanisms by which this connection might exist.

Table 4.45: Teachers integrating STEM methods improve student learning in non-STEM subjects.

Level	Frequency	Percent
Strongly disagree	10	3.3
Disagree	13	4.3
Neutral	58	19.3
Agree	154	51.3
Strongly agree	57	19.0
Total	300	100.0

The data leans towards a positive association between teachers integrating STEM methods and improving student learning in non-STEM subjects. Over two-thirds (69.3%, including those who strongly agree) of respondents believe that incorporating STEM approaches benefits learning in non-STEM areas. Less than 8% (7.6%, including disagree and strongly disagree) disagreed.

This suggests that STEM integration has the potential to be a valuable teaching strategy for a broader range of subjects.

Table 4.46: Integrating STEM principles makes learning in subjects like arts or Music more engaging

Level	Frequency	Percent
Disagree	18	6.0
Neutral	69	23.0
Agree	140	46.7
Strongly agree	61	20.3
Total	300	100.0

The data suggests that integrating STEM principles can make learning in arts or music more engaging for students. Over two-thirds (67.0%, including those who strongly agree) believe STEM integration enhances engagement in these subjects. A relatively small percentage (29.0%, including disagree and neutral) disagreed or were unsure about the benefits. This indicates that STEM integration has the potential to be a promising strategy for increasing student interest and motivation in traditionally non-STEM fields.

Table 4.47: It is feasible to integrate STEM activities into non-STEM subjects such as arts, humanities, and social sciences.

Level	Frequency	Percent
Strongly disagree	7	2.3
Disagree	10	3.3
Neutral	69	23.0
Agree	150	50.0
Strongly agree	58	19.3
Total	300	100.0

The data strongly suggests that integrating STEM activities into non-STEM subjects like arts, humanities, and social sciences is feasible. Over two-thirds (69.3%, including those who strongly agree) believe STEM activities can be successfully incorporated into these subjects.

Less than 6% (5.6%, including disagree and strongly disagree) disagreed. This high level of agreement indicates that there is significant potential for cross-disciplinary learning through STEM integration. However, a notable portion (23.0%) remained neutral.

Table 4.48: Integrating STEM enhances critical thinking skills in students studying subjects beyond traditional STEM fields

Level	Frequency	Percent
Strongly disagree	5	1.7
Disagree	13	4.3
Neutral	56	18.7
Agree	151	50.3
Strongly agree	69	23.0
Total	300	100.0

The data strongly supports the idea that integrating STEM education can enhance critical thinking skills in students across all disciplines. Over three-quarters (73.3%, including those who strongly agree) of respondents believe STEM integration benefits critical thinking in non-STEM fields. Only a small percentage (6.0%, including disagree and strongly disagree) disagreed. This suggests that STEM integration has the potential to be a valuable method for developing critical thinking skills, a foundational ability applicable to a wide range of subjects.

Table 4.49: It is challenging for educators to effectively integrate STEM into non-STEM subjects like Fine Arts or History

Level	Frequency	Percent
Strongly disagree	7	2.3
Disagree	20	6.7

Neutral	68	22.7
Agree	135	45.0
Strongly agree	62	20.7
Total	300	100.0

The data suggests that integrating STEM into non-STEM subjects like Fine Arts or History presents challenges for educators, but there is still potential. While a significant portion (65.7%, including those who strongly agree) believe teacher integration of STEM in non-STEM subjects is challenging, a notable group (29.0%, including disagree and strongly disagree) doesn't necessarily see it as such. This highlights a potential obstacle in implementing STEM integration across disciplines.

Table 4.50: The project of art and design incorporates principles of science, technology, engineering, or math (STEM) into the creative design process

Level	Frequency	Percent
Strongly disagree	2	.7
Disagree	19	6.3
Neutral	67	22.3
Agree	151	50.3
Strongly agree	53	17.7
total	300	100.0

The data results reveal a strong connection between STEM principles and the creative process in art and design. Over two-thirds of respondents (agree + strongly agree) believe STEM plays a role in artistic endeavors. This is further supported by the minimal disagreement (7%). While a sizeable portion (22.3%) remained neutral, the significant positive skew towards agreement suggests a widespread recognition of STEM's influence on artistic and design thinking.

Table 4.51: STEM activities can make subjects like arts and humanities more engaging and interesting for students

Level	Frequency	Percent
Disagree	10	3.3
Neutral	55	18.3
Agree	150	50.0
Strongly agree	69	23.0
Total	300	100.0

The data suggests a promising connection between STEM activities and increased engagement in arts and humanities subjects. A strong majority (73%) of respondents agreed that STEM activities can make these subjects more interesting. This positive sentiment is further bolstered by the minimal disagreement (3.3%). While nearly one-fifth (18.3%) remained neutral, the significant positive skew

towards agreement indicates that many see STEM activities as a potential tool to boost engagement in traditionally non-STEM fields.

Table 4.52: Integrating STEM activities in non-STEM courses helps students develop practical skills applicable to their field of study

Level	Frequency	Percent
Strongly disagree	6	2.0
Disagree	10	3.3
Neutral	53	17.7
Agree	151	50.3
Strongly agree	75	25.0
total	300	100.0

The survey results strongly support the integration of STEM activities in non-STEM courses for developing practical skills. A resounding three-quarters of respondents (agree + strongly agree) believe such integration benefits students. This is further bolstered by the minimal disagreement (5.3%). While some remained neutral (17.7%), the significant positive skew towards agreement suggests a widespread recognition of STEM activities' potential to equip students with practical skills valuable in their chosen fields, regardless of being traditionally STEM-related.

Table 4.53: Do you think Completing multiple STEM projects helps students develop essential skills such as critical thinking and problem-solving.

Level	Frequency	Percent
Strongly disagree	4	1.3
Disagree	15	5.0
Neutral	53	17.7
Agree	140	46.7
Strongly agree	78	26.0
Total	300	100.0

The data overwhelmingly supports the notion that engaging in multiple STEM projects fosters critical thinking and problem-solving skills in students. Nearly three-quarters (72.7%) of respondents agreed that such projects are beneficial, with a significant portion (26%) strongly agreeing. This positive sentiment is further bolstered by the minimal disagreement (6.3%). While some remained neutral (17.7%), the significant positive skew towards agreement suggests a broad consensus that completing STEM projects equips students with essential critical thinking and problem-solving abilities.

Table 4.54: Stem activities like Constructing bridges, towers, or other structures using materials like straws, popsicle sticks can help students to develop creativity skills.

Level	Frequency	Percent
Strongly disagree	5	1.7
Disagree	21	7.0
Neutral	52	17.3
Agree	151	50.3
Strongly agree	64	21.3
Total	300	100.0

The data results provide compelling evidence that hands-on STEM activities like building structures with straws and popsicle sticks nurture creativity in students. Over two-thirds of respondents (agree + strongly agree) believe these activities promote creative thinking. This is further strengthened by the minimal disagreement (8.7%). While a sizeable portion (17.3%) remained neutral, the significant positive skew towards agreement suggests a widespread recognition of such STEM activities as valuable tools for fostering creativity in students.

Table 4.55: Do you think stem activities used in science concepts such as physics help students to understand the concept better.

Level	Frequency	Percent
Strongly disagree	4	1.3
Disagree	15	5.0
Neutral	53	17.7
Agree	140	46.7
Strongly agree	78	26.0
Total	300	100.0

The data results overwhelmingly endorse the use of STEM activities to enhance student understanding of science concepts like physics. Nearly three-quarters (72.7%) of respondents agreed that such activities are beneficial, with a significant portion (26%) strongly agreeing. This positive sentiment is bolstered by the minimal disagreement (6.3%). While some remained neutral (17.7%), the significant positive skew towards agreement suggests a broad consensus that STEM activities in physics education act as powerful tools for improving student comprehension.

4.7 Variable wise Sum of Responses

Table 4.56: Perceptions of STEM Integration to other Subjects

	Frequency	Percent
Low (≤ 12.00)	1	.3

Medium (13.00 - 36.00)	29	9.7
High (37.00+)	266	88.7
Total	296	98.7

This table reveals a strong positive perception towards STEM integration among the surveyed population. The vast majority, with 266 responses (88.7%), fall under the "High (37.00+)" category. This suggests a significant portion of respondents perceive STEM integration very favorably. While a smaller group (29 respondents, 9.7%) falls within the "Medium (13.00 - 36.00)" range, indicating a somewhat positive to neutral perception, only a single respondent (0.3%) falls under "Low (≤ 12.00)," signifying very low or negative perception. Overall, the data demonstrates a clear trend of positive sentiment regarding STEM integration with other subjects.

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Figure 4.1: STEM Integration to other subjects (Pie Chart)

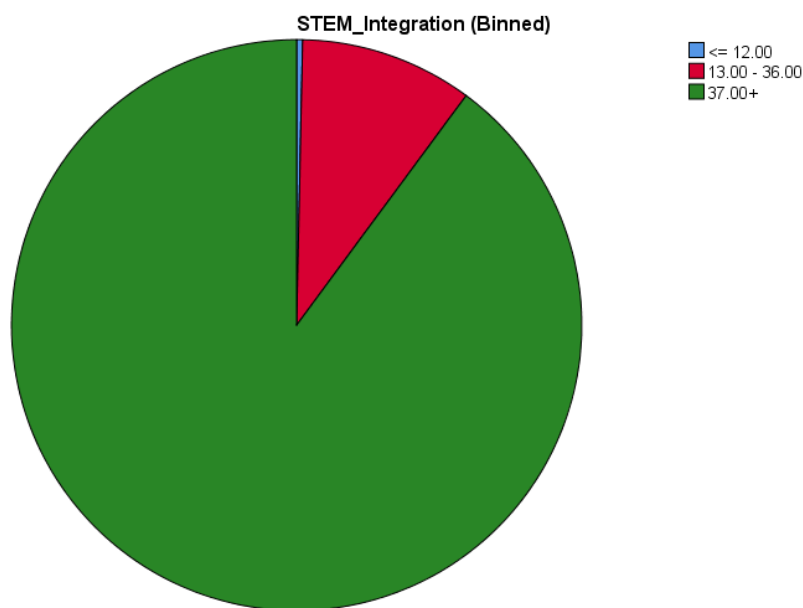


Table 4.57: Role of STEM in enhancing academic performance

	Frequency	Percent
Low (≤ 10.00)	1	.3
Medium (11.00 - 30.00)	31	10.3
high (31.00+)	266	88.7
Total	300	100.0

The data in this table overwhelmingly suggests a positive perception of STEM's influence on academic performance. A substantial majority of respondents, at 266 (88.7%), placed the role of STEM in the "high" category. This indicates a strong belief that STEM education is highly beneficial for academic

achievement. An additional 31 respondents (10.3%) categorized STEM's role as "medium," suggesting a positive or neutral perception. With only a single respondent (0.3%) falling under "low," there is negligible evidence of a negative view on STEM's impact. Overall, the findings provide strong support for the argument that STEM education plays a significant role in enhancing academic performance.

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Figure 4.2: STEM and Academic Performance

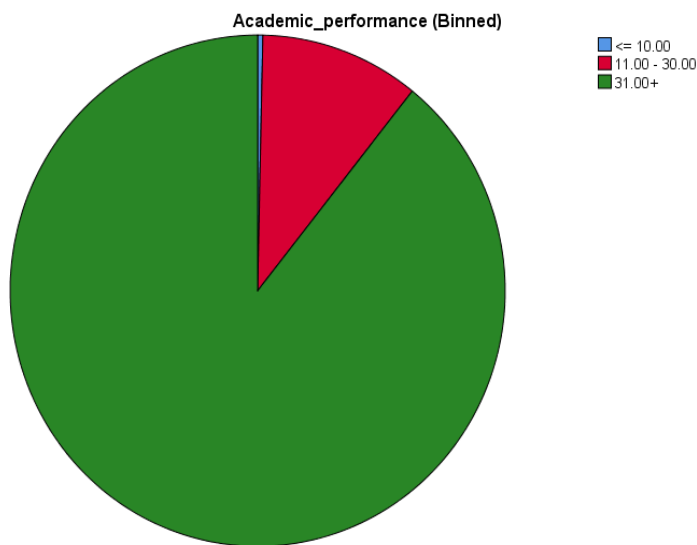


Table 4.58: Role of STEM in boosting Problem Solving skills

		Frequency	Percent
Low	(<= 7.00)	1	.3
Medium	(8.00 - 23.50)	29	9.7
High	(23.51+)	267	89.0
Total		300	100.0

The data in this table overwhelmingly supports a positive correlation between STEM education and enhanced problem-solving abilities. A striking majority, consisting of 267 respondents (89.0%), placed STEM's role in the "High" category. This signifies a widespread belief that STEM education significantly strengthens problem-solving skills. An additional 29 respondents (9.7%) categorized STEM's contribution as "Medium," indicating a positive or neutral perception. With a mere 1 respondent (0.3%) falling under "Low," there is minimal evidence to suggest a negative view on STEM's impact on problem-solving skills. In conclusion, the findings provide compelling support for the argument that STEM education plays a crucial role in developing problem-solving abilities.

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Figure 4.3: STEM and problem-solving skills

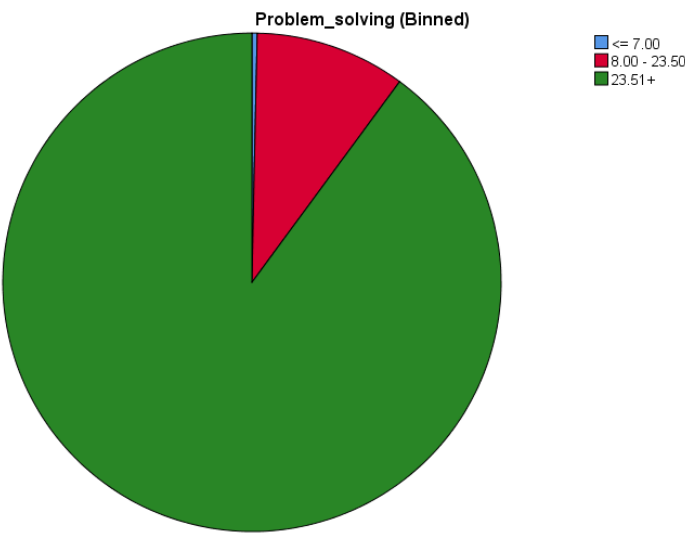
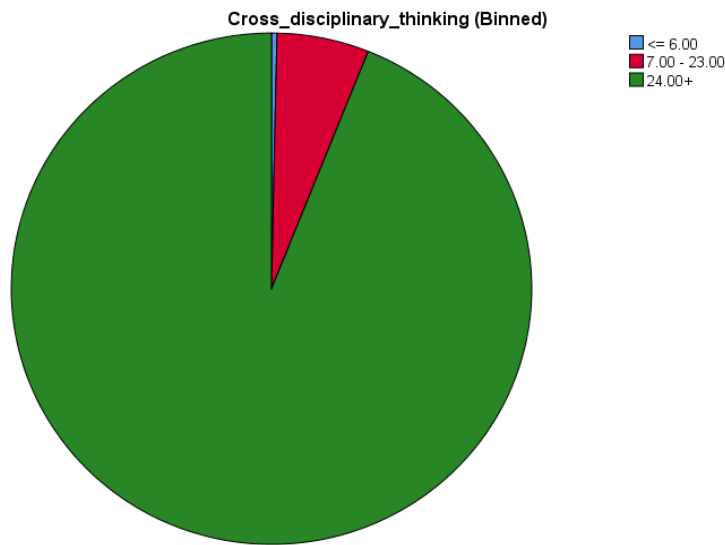


Table 4.59: STEM and Cross-disciplinary thinking

		Frequency	Percent
Low	(<= 6.00)	1	.3
Medium	(7.00 - 23.00)	17	5.7
High	(24.00+)	279	93.0
Total		300	100.0

The data presented in this table overwhelmingly highlights the positive perception of STEM education's influence on developing cross-disciplinary thinking skills. A remarkable 279 respondents (93.0%) placed STEM's role in the "High" category. This signifies a widespread belief that STEM education significantly fosters the ability to apply knowledge and connect concepts across different disciplines. An additional 17 respondents (5.7%) categorized STEM's contribution as "Medium," indicating a positive or neutral perception. With only a negligible number of respondents (1 or 0.3%) falling under "Low," there is minimal evidence of a negative view on STEM's impact. Overall, the findings provide strong support for the argument that STEM education plays a crucial role in developing cross-disciplinary thinking abilities.

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Figure 4.4: STEM and cross disciplinary thinking**Table 4.60: STEM and Practical application learning**

	Frequency	Percent
Low (<= 6.00)	1	.3
Medium (7.00 - 22.50)	26	8.7
High (22.51+)	270	90.0
Total	300	100.0

The data in this table overwhelmingly supports a positive correlation between STEM education and practical application learning. A striking majority, consisting of 270 respondents (90.0%), placed STEM's role in the "High" category. This signifies a widespread belief that STEM education significantly strengthens practical application learning skills. An additional 26 respondents (8.7%) categorized STEM's contribution as "Medium," indicating a positive or neutral perception. With a mere 1 respondent (0.3%) falling under "Low," there is minimal evidence to suggest a negative view on STEM's impact on practical application learning. In conclusion, the findings provide compelling support for the argument that STEM education plays a crucial role in developing practical application learning abilities.

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Figure 4.5: STEM and practical life applications

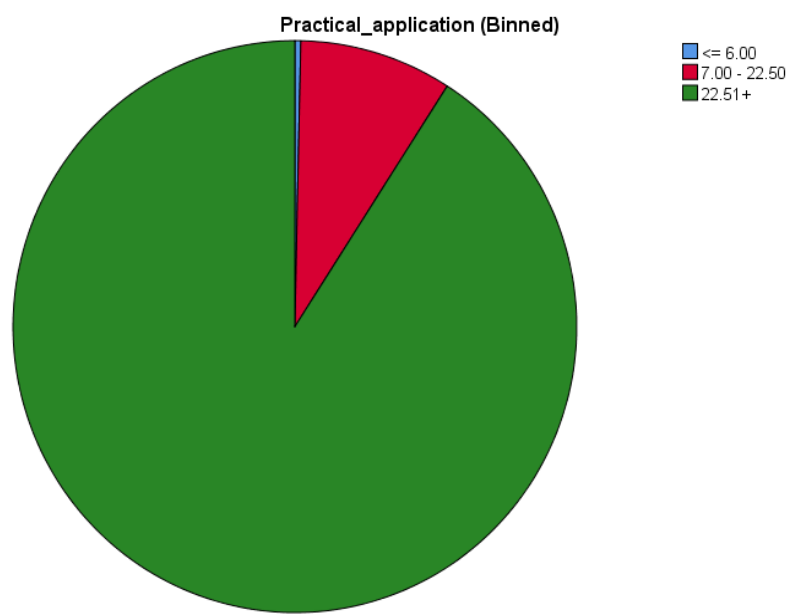
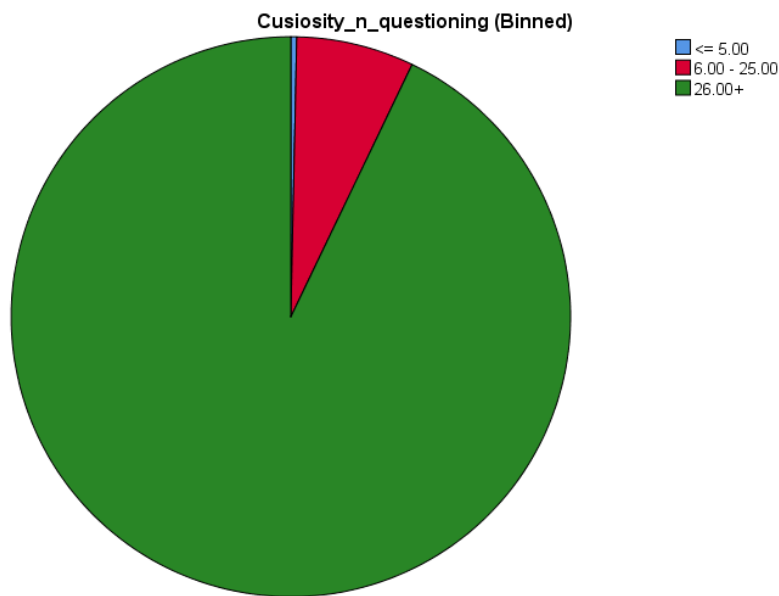


Table 4.61: STEM in raising curiosity

		Frequency	Percent
Low	(<= 5.00)	1	.3
Medium	(6.00 - 25.00)	20	6.7
High	(26.00+)	273	91.0
Total		300	100.0

The data in this table overwhelmingly supports a positive correlation between STEM education and raising curiosity. A striking majority, consisting of 273 respondents (91.0%), placed STEM's role in the "High" category. This signifies a widespread belief that STEM education significantly strengthens curiosity skills. An additional 20 respondents (6.7%) categorized STEM's contribution as "Medium," indicating a positive or neutral perception. With a mere 1 respondent (0.3%) falling under "Low," there is minimal evidence to suggest a negative view on STEM's impact on raising curiosity. In conclusion, the findings provide compelling support for the argument that STEM education plays a crucial role in developing curiosity abilities.

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Figure 4.6: STEM and curiosity and thinking**Table 4.62: Impact of STEM on Creative thinking skills**

	Sum of Squares	Mean Square	F	Sig.
Academic performance	5.298	2.649	29.823	.000
Problem solving	5.467	2.734	32.881	.000
Cross disciplinary thinking	4.137	2.069	38.743	.000
Practical application	6.651	3.326	47.073	.000
Curiosity and questioning	4.487	4.487	73.332	.000

This table provides compelling evidence for a statistically significant positive relationship between STEM education and various aspects of creative thinking. The highly significant F-values (all exceeding .001) for "Academic Performance," "Problem Solving," "Cross-disciplinary Thinking," "Practical Application," and "Curiosity and questioning" all reject the null hypothesis of no association. This means there's a strong likelihood that STEM education has a positive impact on these creative thinking skills.

Looking deeper, the high mean squares for each category suggest a substantial influence of STEM on these creative thinking skills. For instance, the highest mean square (4.487) for "Curiosity and questioning" indicates a potentially stronger positive effect of STEM on this specific aspect of creative thinking compared to others.

Overall, the data presented in this table strongly supports the claim that STEM education fosters various creative thinking skills, potentially leading to improved academic performance, problem-

solving abilities, cross-disciplinary thinking, practical application of knowledge, and a heightened sense of curiosity and questioning.

4.8 Summary

Overall, the data suggests a positive correlation between STEM education and creativity. Studies show a positive link between STEM education and creativity. Students in STEM programs report stronger creative problem-solving skills, a belief in creativity's importance for STEM success, and an enhanced ability to think critically and approach problems from new angles. Notably, hands-on, project-based learning seems particularly effective in nurturing creativity. Students involved in such activities feel that STEM education fosters creative thinking. However, there's room for improvement in integrating cross-disciplinary thinking. While some see the connection between STEM fields, many remain neutral or disagree. Overall, these findings highlight the importance of incorporating creative thinking and problem-solving into STEM education. This can be achieved through encouraging students to ask questions and develop hypotheses, providing opportunities for hands-on learning and experimentation, integrating project-based learning that tackles real-world problems, and fostering collaboration and teamwork among students.

Chapter 5

SUMMARY, FINDINGS, CONCLUSION, DISCUSSION AND RECOMMENDATIONS

This chapter presents the summary, findings, conclusion, discussion and recommendations of the study.

5.1 Summary

This study investigated the factors influencing student creativity within STEM education programs. The primary objective was to assess how STEM education approaches students' creative thinking skills.

The target population for this study included students enrolled in STEM programs at public universities across Lahore. A descriptive research design was employed, utilizing a survey method. A convenient sampling approach was used to select student participants. To gather data, a researcher-developed questionnaire with 55 items was administered to students currently enrolled in BS programs. Following a rigorous evaluation of the instrument's validity and reliability, data collection was conducted by the researcher directly from the target population. Upon data collection, SPSS software was employed for statistical analysis, generating percentages and descriptive statistics to interpret the results. The analysis led to findings and conclusions, which subsequently informed recommendations for future research and implementation of STEM education practices that nurture student creativity

5.2 Findings

On the basis of interpretation of data following findings are given below.

(Question number 1-10 relates to academic performance)

1. Over half of the respondents agreed that STEM education encourages students to think creatively in problem-solving activities.
2. A large majority (nearly 80%) of respondents agreed that STEM education improves problem-solving skills, which benefits academic performance.

3. Opinions are mixed on STEM education's impact on all subjects. While a majority agrees (59.7%), a substantial portion (30.4%) is neutral or disagrees.
 4. Creativity is seen as vital for STEM success. Over two-thirds (66.6%) of respondents agreed that creativity is important, suggesting a shift beyond rote memorization in STEM education.
 5. Strong support (66%) exists for STEM projects fostering creativity in students. There's minimal disagreement (10%), suggesting project-based learning is a valuable tool for nurturing creativity in STEM education.
 6. Almost 70% of respondents believe creativity helps in getting good grades in STEM courses, highlighting its importance in STEM education.
 7. Opinions are divided on STEM's interdisciplinary approach fostering creativity (54% agree, 40.3% neutral/disagree), suggesting curriculum design and implementation might be key factors.
 8. Majority (66.3% agree + strongly agree) believe STEM education fosters creative problem-solving skills that benefit academic achievement.
 9. Majority (67% agree + strongly agree) say STEM education fosters critical thinking skills that benefit academic performance.
 10. Majority (65.6% agree + strongly agree) believe STEM education fosters curiosity and creativity, suggesting it sparks a love for learning that boosts creative thinking across subjects.
- (Question 11-18 relates to problem solving and innovation)
11. Most of the students (68.7% agree + strongly agree) believe STEM education fosters creative problem-solving, highlighting its value in equipping students to think creatively.
 12. Most of the respondents (62% agree + strongly agree) believe problem-solving in STEM promotes innovative thinking, but there's room for improvement (33.3% disagree).
 13. Majority (64.3% agree + strongly agree) believe STEM education fosters innovation and adaptability, key skills for the modern world.

14. Nearly eight in ten (79.0%) believe creative problem-solving is valuable for STEM success (only 10.7% disagree).
15. Nearly seven in ten (69.3% agree) believe STEM subjects challenge students to develop creative solutions (only 6.0% disagree).
16. Majority (67% agree + strongly agree) believe STEM problem-solving encourages exploring unconventional ideas (only 8% disagree).
17. Majority (68% agree) believe creativity is crucial for problem-solving in STEM subjects (only 6.7% disagree).
18. Majority (66.7% agree + strongly agree) believe STEM students collaborate to solve real-world problems creatively (only 6.3% disagree), highlighting the value of collaborative learning in STEM education.

(Question 19-26 relates to the cross disciplinary thinking)

19. Most of the respondents (60% agree + strongly agree) believe STEM education fosters curiosity and creativity (20.7% disagree), suggesting it sparks a love for learning that boosts creative thinking across subjects.
20. Most of the respondents (67% agree + strongly agree) believe cross-disciplinary thinking is important for solving problems in STEM (only 8% disagree).
21. Majority (68.7% agree + strongly agree) believe STEM education integrates knowledge from various disciplines (only 7.7% disagree), highlighting the importance of making connections between different subjects in STEM.
22. Majority (73% agree + strongly agree) believe exploring connections between STEM fields boosts creativity (only 5% disagree).
23. Majority (70.7% agree + strongly agree) believe cross-disciplinary thinking is key for creative problem-solving in STEM (only 6% disagree).

24. Most of the respondents (72.7% agree + strongly agree) believe STEM education integrates knowledge across disciplines for solving complex problems.
25. Mostly respondents (71.4% agree + strongly agree) believe STEM education fosters a multifaceted approach to learning.
26. Nearly seven in ten (68.7% agree + strongly agree) believe creativity is a valuable asset for STEM success.

(Questions 27-34 relates to practical application)

27. Over half (54.0% agree) believe STEM education promotes creative thinking through hands-on experiences, with a significant portion (25.3%) neutral and a small minority disagreeing (17.7%).
28. Over two-thirds (70.4% agree) believe applying STEM knowledge enhances creative problem-solving skills
29. Nearly eight in ten (79.7% agree) believe STEM education fosters creativity through real-world application (only 6% disagree).
30. Nearly two-thirds (68.7% agree) of respondents believe fieldwork and experiments enhance understanding of STEM subjects, with minimal disagreement (6.7%).
31. Majority (69.3%) agree STEM education fosters creative problem-solving (weak disagreement: 8.0%).
32. Majority (69.3%) believe STEM education challenges students to solve real-world problems creatively (weak disagreement: 8.0%).
33. Majority (72.3% agree) believe practical STEM experiences foster innovative thinking and adaptability.
34. Mostly respondents (65.7% agree) believe applying STEM knowledge helps explore creative solutions for real-world problems (7% disagree), but a significant portion (24.3%) remained neutral.

(Question 35-43 relates to the curiosity and questioning)

35. Most of the respondents (62% agree) believe STEM education fosters curiosity and questioning (minimal disagreement: 8%), with a significant neutral group (26%).
36. Nearly seven in ten (68.7% agree) believe curiosity and questioning are important for STEM creativity.
37. Majority (70.3% agree) believe STEM education fosters curiosity and exploration.
38. Majority (68.7% agree) believe STEM education fosters curiosity (very small disagreement: 7.7%), but a significant portion (19.7%) remained neutral.
39. Majority (68.7% agree) believe curiosity in STEM sparks creativity (very small disagreement: 6.0%).
40. Mostly respondents (65% agree) find STEM students inquisitive about 'why' and 'how' (neutral: 25.7%, very small disagreement: 1.7%), suggesting STEM fosters curiosity but room for improvement exists.
41. Over 60% see a possible link between STEM and arts/literature understanding, but a significant portion (36.1%) is unsure (disagree or neutral), suggesting more research is needed.
42. Majority (73.4% agree) believe STEM education fosters an investigative learning approach (very small disagreement: 5.0%).
43. Majority (71% agree) believe STEM education improves critical thinking skills (less than 6% disagree), but a significant portion (20.0%) remained neutral.

(Question 44-55 relates to the perception of stem integration into NON-STEM subjects)

44. Over 60% see a possible link between STEM and arts/literature understanding, but a significant portion (36.1%) is unsure (disagree or neutral), suggesting more research is needed.
45. Over two-thirds (69.3% agree) believe STEM methods improve learning in non-STEM subjects.
46. Majority (67% agree) believe STEM integration boosts engagement in arts or music.

47. Majority (69.3% agree) believe STEM activities can be integrated into non-STEM subjects (less than 6% disagree), but a notable portion (23.0%) remained neutral.
48. Most of the respondents (73.3% agree) believe STEM integration benefits critical thinking across disciplines.
49. Majority (65.7% agree) find integrating STEM into non-STEM subjects challenging, but a notable portion (29.0%) disagrees, highlighting a potential obstacle.
50. Mostly respondents (67.3% agree + strongly agree) believe STEM principles influence art and design, with minimal disagreement (7.0%).
51. Mostly respondents (73% agree) believe STEM activities can make arts and humanities more engaging (minimal disagreement: 3.3%), suggesting they may be a tool to boost student interest.
52. Majority (72.7% agree) believe STEM projects foster critical thinking and problem-solving skills
53. Majority (75% agree) believe integrating STEM activities in non-STEM courses benefits students (minimal disagreement: 5.3%), highlighting its potential to develop practical skills.
54. Most of the respondents (66.3% agree) believe hands-on STEM activities like building with straws foster creativity (minimal disagreement: 8.7%).
55. Majority (72.7% agree) believe STEM activities improve understanding of science concepts (minimal disagreement: 6.3%).

5.3 Discussions

STEM education can act as a springboard for igniting creativity in students. When STEM classes are engaging and interactive, they can spark a passion for learning that translates into higher participation and a more positive learning environment. This intrinsically motivating atmosphere can indirectly benefit students' mental health by fostering a sense of enjoyment and accomplishment. STEM education also equips students with valuable tools for navigating challenges and managing stress. By

encouraging critical thinking and problem-solving skills, students develop a strategic approach to overcoming obstacles, fostering resilience and a sense of control. Furthermore, successfully navigating complex problems and completing STEM projects can significantly boost students' confidence and self-esteem. The sense of accomplishment gained from overcoming challenges fuels a positive self-image and a belief in one's own abilities. STEM education isn't a solitary pursuit either. Many STEM fields heavily emphasize teamwork and collaboration. Working together on projects fosters a sense of community and belonging for students. This social aspect can be particularly important for mitigating feelings of isolation, which can negatively impact mental health. Finally, STEM education often champions experimentation, iteration, and learning from mistakes. This approach cultivates a growth mindset in students, where they view challenges as opportunities to learn and improve rather than insurmountable roadblocks. By fostering this adaptable and solution-oriented approach, STEM education can empower students to approach problems creatively and think outside the box.

5.4 Conclusion

In conclusion, the data overwhelmingly supports the positive impact of STEM education on fostering creativity in students. This is evident across several key areas. Firstly, engaging and well-designed STEM programs can spark a passion for learning, leading to increased participation and a more positive classroom environment. This intrinsically motivating atmosphere fosters a sense of enjoyment and accomplishment, which can indirectly benefit students' overall well-being and mental health. Secondly, STEM education equips students with essential tools for critical thinking and problem-solving. By encouraging them to analyze situations, develop solutions, and experiment with different approaches, STEM classrooms become training grounds for creative thinking. Overcoming challenges and successfully completing STEM projects further fuels this creativity by boosting students' confidence and self-esteem. The sense of accomplishment gained from navigating complex problems fosters a belief in one's own abilities and a willingness to take creative risks. Furthermore, STEM fields often emphasize collaboration and teamwork. Working together on projects fosters a sense of community

and belonging, providing a supportive network for students to brainstorm ideas, share perspectives, and build upon each other's strengths. This collaborative environment is a breeding ground for creative solutions and innovative thinking. Finally, STEM education inherently promotes a growth mindset. By encouraging experimentation, embracing iteration, and learning from mistakes, STEM classrooms challenge students to view obstacles as opportunities for growth and development. This adaptability and solution-oriented approach empowers students to think outside the box and approach problems with a creative lens. In essence, STEM education equips students with the skills, confidence, and collaborative spirit necessary to thrive in a world that demands creativity and innovation.

5.5 Recommendations

1. Allocate resources to develop engaging and interactive STEM programs that spark curiosity and a passion for learning. This intrinsically motivating environment fosters a willingness to explore and experiment, key aspects of creative thinking.
2. Implement project-based learning experiences that encourage critical thinking, problem-solving, and collaboration. By tackling real-world challenges and developing innovative solutions, students hone their creative approach.
3. Incorporate design thinking principles into STEM classrooms. This framework emphasizes user empathy, brainstorming diverse solutions, prototyping, and iterating on ideas. This process fosters a creative approach to problem-solving and encourages students to think outside the box.
4. Break down silos between STEM disciplines and encourage collaboration across fields. This allows students to combine diverse perspectives and skillsets, fostering a more multifaceted approach to problem-solving, which is essential for creative endeavors.
5. Designate time for open-ended exploration within STEM classes. This allows students to experiment with different approaches, make mistakes, and learn from them, fostering a growth mindset and a willingness to take creative risks.

6. Offer professional development workshops for STEM instructors, equipping them with strategies to nurture creativity in the classroom. This may include techniques like open-ended questioning, encouraging student inquiry, and valuing diverse perspectives.

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APPENDIX

Impact of stem education on creativity of university students

Gender: Male ☐ female ☐

Institute name: _____

Education level: Under graduate ☐ Post graduate ☐

Age: 18-22 ☐ 22-26 ☐

Directions: For each statement in a survey, please indicate how much you **agree** or **disagree** with the statement on the right side of each statement.

Scale: 5- Strongly agree

4- Agree

3- Neutral

2- Disagree

1- Strongly disagree

Academic performance

	5	4	3	2	1
1. STEM education encourages students to think creatively in problem-solving activities.					
2. STEM education has enhanced problem-solving skills, which has a positive impact on overall academic performance					
3. STEM education has a positive influence on overall academic performance across all subjects.					
4. Creativity plays a significant role in academic performance in STEM education.					
5 STEM projects and hands-on activities enhance students' creative thinking abilities.					
6. Creativity is a valuable asset for achieving higher grades in STEM courses					
7. STEM's interdisciplinary approach encourages creative connections between different subjects					
8. Creative problem-solving skills acquired in STEM positively impact overall academic achievement.					

9. Critical thinking skills have improved as a result of participation in STEM education, leading to better academic outcomes					
10. STEM education fosters a culture of curiosity, which enhances academic creativity.					
Problem solving and innovation	5	4	3	2	1
1.STEM education encourages students to approach problems creatively					
2. Problem-solving in STEM subjects promotes innovative thinking in students.					
3. STEM education fosters a culture of innovation and adaptability in students.					
4. Creative problem-solving is a valuable skill for success in STEM fields.					
5. STEM subjects challenge students to find creative solutions to complex challenges					
6. Problem-solving activities in STEM encourage students to explore unconventional ideas.					
7. Creativity plays a crucial role in problem-solving in STEM subjects.					
8. STEM students often collaborate to devise creative solutions to real-world problems					

Cross disciplinary thinking	5	4	3	2	1
1. STEM education fosters a culture of curiosity, which enhances academic creativity.					
2. Cross-disciplinary thinking is a key element in the problem-solving process in STEM subjects.					

3. STEM education challenges students to integrate knowledge from various disciplines					
4. Creativity is enhanced when students explore the intersections between different fields within STEM.					
5. Cross-disciplinary thinking promotes creative problem-solving in STEM subjects.					
6. STEM education has helped to integrate concepts from different disciplines to solve complex problems					
7. Interdisciplinary connections in STEM contribute to students' ability to approach problems from innovative angles.					
8. Creativity plays a significant role in success in STEM fields that demand cross-disciplinary thinking.					

Practical application	5	4	3	2	1
1. STEM education promotes creative thinking by providing practical, hands-on experiences.					
2. Practical application of knowledge in STEM subjects enhances students' creative problem-solving skills.					
3. STEM subjects encourage students to apply theoretical knowledge to real-world scenarios, fostering creativity.					
4. Engaging in fieldwork and experiments in STEM education has provided students with practical experience that enhances their understanding of the subject matter.					
5. Creativity is essential when students work on practical, real-world applications in STEM education.					
6. STEM subjects challenge students to find creative solutions to practical challenges they encounter.					
7. Practical experience gained through STEM education leads to innovative thinking and adaptability.					
8. The application of knowledge in STEM enhances students' ability to explore creative solutions in real-world contexts.					

Curiosity and Questioning	5	4	3	2	1
1.STEM education encourages students to be curious and ask questions					
2. Curiosity and questioning are important drivers of creativity in STEM subjects					
3. STEM subjects foster a culture of curiosity, inspiring students to explore new ideas.					
4. STEM education has significantly increased student's curiosity about how things work in the world around them					
5. Creative thinking is often sparked by students' curiosity about STEM topics.					
6. STEM students frequently inquire about the "why" and "how" behind scientific and technological phenomena					
7. STEM subjects challenge students to find creative solutions to practical challenges they encounter.					
8. STEM education has encouraged students to adopt an investigative approach to learning, where they actively seek out information and answers.					
9. student ability to critically question and evaluate information has improved due to their involvement in STEM education					

Perceptions of STEM Integration in Non-STEM Subjects	5	4	3	2	1
1. STEM concepts enhance understanding in subjects like Fine Arts or Literature.					
2. Teachers integrating STEM methods improve student learning in non-STEM subjects.					
3. Integrating STEM principles makes learning in subjects like arts or Music more engaging.					
4. It is feasible to integrate STEM activities into non-STEM subjects such as arts, humanities, and social sciences.					

5. Integrating STEM enhances critical thinking skills in students studying subjects beyond traditional STEM fields.					
6. It is challenging for educators to effectively integrate STEM into non-STEM subjects like Fine Arts or History.					
7. The project of art and design incorporates principles of science, technology, engineering, or math (STEM) into the creative design process.					
8. STEM activities can make subjects like arts and humanities more engaging and interesting for students					
9. Integrating STEM activities in non-STEM courses helps students develop practical skills applicable to their field of study					
10. do you think Completing multiple STEM projects helps students develop essential skills such as critical thinking and problem-solving.					
11. stem activities like Constructing bridges, towers, or other structures using materials like straws, popsicle sticks can help students to develop creativity skills.					
12. do you think stem activities used in science concepts such as physics help students to understand the concept better.					

